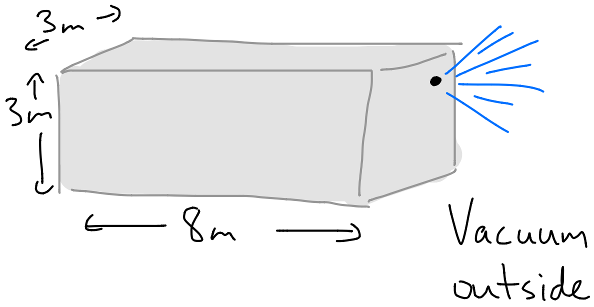
University Physics A(2) 2014

Worksheet #11: Gases and Engines

Name (名字):   Student number (学号):

Problems Show all working.

(1) [M&I.13.P.21] You are on a spacecraft measuring 8 m by 3 m by 3 m when it is struck by a piece of space junk (垃圾), leaving a circular hole of radius 4 mm. About how much time do you have to repair the hole? Explain whatever approximations you make.



Assume conditions inside the spacecraft are initially STP.

$\rightarrow$ $n_0 = 2.7 \times 10^{25}$ molecules/m³. ← (I looked this up in the book!)

and $N_0 = n_0 V = (2.7 \times 10^{25})(8)(3)(3) = 1.9 \times 10^{27}$ molecules

Mass of one molecule: (for N_2)

$$m = \frac{0.028 \text{ kg/mol}}{6.02 \times 10^{23} \text{ molecules/mol}} = 4.65 \times 10^{-26} \text{ kg}$$

$$\rightarrow \bar{v} \simeq v_{rms} = \sqrt{\frac{3k_B T}{m}} = \sqrt{\frac{3(1.4 \times 10^{-23})(273)}{4.65 \times 10^{-26}}} \simeq 500 \text{ m/s}$$

Leakage rate $= \frac{1}{4} n A \bar{v}$ [molecules/s]

① A very rough approximation:

$\boxed{\text{Assume the leakage rate is constant.}}$

$$\rightarrow \text{Rate} = \frac{1}{4} n_0 A \bar{v} = \frac{1}{4}(2.7\times10^{25})(\pi.[4\times10^{-3}]^2)(500)$$

$$= 1.7\times10^{23} \text{ molecules/s}$$

$$\therefore \text{Time to empty} \simeq \frac{N_0}{\text{rate}} = \frac{1.9\times10^{27}\text{ molecules}}{1.7\times10^{23}\,\frac{\text{molecules}}{s}} \simeq \boxed{1\times10^4\,s}$$

(3 hours).

This is good enough - but we can do better! Actually, n is changing as the air leaks out. We can take this into account:

② Less rough approximation:

$\boxed{\text{Leakage rate \underline{not} constant; still assume } T \text{ (and } \bar{v}) \text{ is constant.}}$

$$\text{Leakage rate} = \frac{1}{4} n A \bar{v}$$

$$-\frac{dN}{dt} = \frac{1}{4}\frac{N}{V} A \bar{v}$$

← A differential equation, which we can solve for $N(t)$.

$$\frac{dN}{N} = -\frac{1}{4}\frac{A\bar{v}}{V}dt$$

Integrate both sides:

$$\int_{N_0}^{N}\left(\frac{1}{N}\right)dN = -\frac{1}{4}\frac{A\bar{v}}{V}\int_{0}^{t}dt$$

(Assuming A, $\bar{v}$, V all constant)

$$\ln N\Big|_{N_0}^{N} = -\frac{A\bar{v}}{4V}t\Big|_{0}^{t}$$

$$\ln N - \ln N_0 = -\frac{A\bar{v}}{4V}(t-0)$$

$$\ln\frac{N}{N_0} = -\frac{A\bar{v}}{4V}t$$

$$\frac{N}{N_0} = e^{-\frac{A\bar{v}}{4V}t}$$

$$\rightarrow \boxed{N(t) = N_0 e^{-\frac{A\bar{v}}{4V}t}}$$

How long does it take for 50% of the air to leak out?

i.e. when $N = 0.5 \times N_0$, $t = ?$

Solving for t:

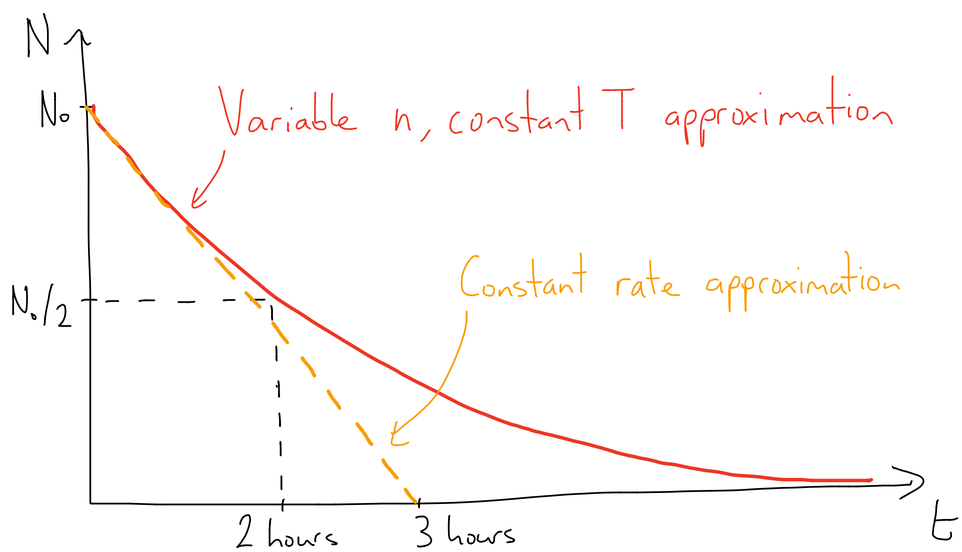
$$t = \frac{4V}{A\bar{v}}\ln\frac{N_0}{N}$$

$$t_{1/2} = \frac{4V}{A\bar{v}} \ln 2$$

$$= \frac{4(8)(3)(3)}{\left(\pi\left[4\times10^{-3}\right]^2\right)(500)} \ln 2$$

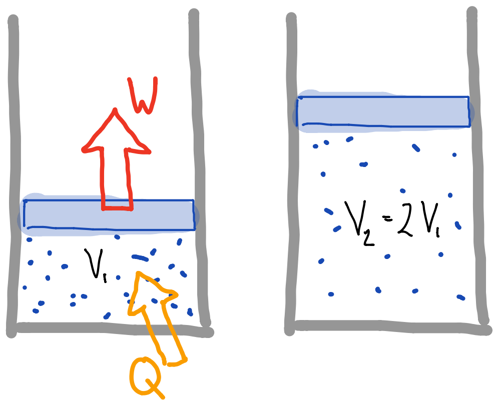
$$= \left(1.1\times10^{4}\,\mathrm{s}\right) \ln 2$$

$$= \boxed{7.6\times10^{3}\,\mathrm{s}} \quad (\sim 2 \text{ hours})$$



(2) (a)[M&I.13.X.24] If you expand the volume of a gas containing N molecules to twice its original volume, while maintaining a constant temperature, how much energy transfer due to a temperature difference is there from the surroundings?

Constant T
→ "isothermal" process



1st Law of Thermodynamics

$$\Delta E = W + Q$$

$$N C_V \cancel{\Delta T}^{\,0} = W + Q$$

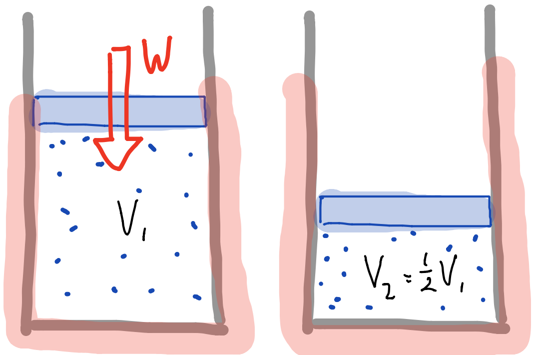
$$\therefore \quad Q = -W_{\text{by surroundings}}$$

$$\rightarrow \quad Q = +\int_{V_1}^{V_2} P\,dV = N k_B T \int_{V_1}^{V_2} \frac{1}{V}\,dV = N k_B T \ln \frac{V_2}{V_1}$$

$$= \boxed{N k_B T \ln 2}$$

(b) [M&I.13.X.25] If you compress a volume of helium containing N atoms to half its original volume in a well-insulated (保温的) container, what is the ratio of the final pressure to the initial pressure? [i.e. P_2/P_1 = ?]

Insulation, so $Q=0$ →



$\boxed{Q=0}$ → Adiabatic process

$$\therefore \quad P_1 V_1^{\gamma} = P_2 V_2^{\gamma}$$

$$\frac{P_2}{P_1} = \left(\frac{V_1}{V_2}\right)^{\gamma}$$

$$= 2^{5/3}$$

$$= \boxed{3.17}$$

Helium (He) is a monatomic gas
(3 degrees of freedom)

$$\rightarrow \quad C_V = 3 \times \frac{1}{2} k_B = \frac{3}{2} k_B$$

$$C_P = C_V + k_B = \frac{5}{2} k_B$$

$$\gamma \equiv \frac{C_P}{C_V} = \frac{\frac{5}{2}k_B}{\frac{3}{2}k_B} = \frac{5}{3}$$

$P_{air} \approx 1 \times 10^5 \text{ Pa}$

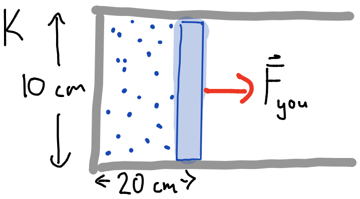
$C_V = \frac{3}{2} k_B$ (monatomic)

(3) [M&I.13.P.27] A horizontal cylinder (圆柱) 10 cm in diameter contains helium gas (He) at room temperature and atmospheric pressure. A piston keeps the gas inside a region of the cylinder 20 cm long.

so, not enough time for energy to escape to the outside

(a) If you *quickly* pull the piston outward a distance of 12 cm, what is the approximate temperature of the helium immediately afterward? What approximations did you make?

$T_1 = 293 \text{ K}$

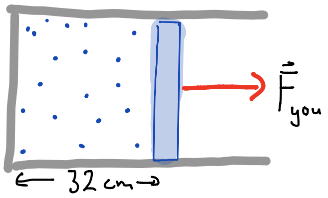


Assume $Q = 0$ (adiabatic)

$$\therefore T_1^{\frac{C_V}{k_B}} V_1 = T_2^{\frac{C_V}{k_B}} V_2$$

$$T_2 = \left(\frac{V_1}{V_2}\right)^{\frac{k_B}{C_V}} T_1$$

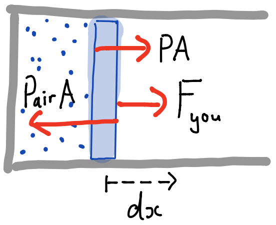
$T_2 = ?$



$$= \left(\frac{20}{32}\right)^{2/3} (293 \text{ K}) = \boxed{214 \text{ K}}$$

(b) How much work did you do, including the sign (i.e. positive or negative)? (HINT: you need to consider the outside air as well as the inside helium.)

A = area of cylinder $= \pi (0.05)^2 = 7.9 \times 10^{-3} \text{ m}^2$



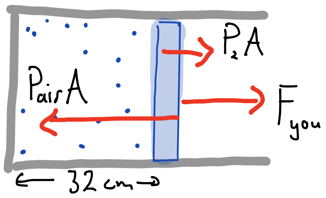
If the expansion is quasistatic, the acceleration of the piston is always ~0, so $\boxed{F_{you} = P_{air}A - PA}$

$$W_{by\ you} = \int F_{you}\, dx = \int [P_{air}A - PA]\, dx = P_{air}\int_{V_1}^{V_2} dV - \int_{V_1}^{V_2} P\, dV$$

$= W_{on\ gas} = N C_V \Delta T = \frac{P_{air} V_1}{k_B T_1} C_V (T_2 - T_1)$

$$= P_{air}(V_2 - V_1) + P_{air} V_1 \frac{C_V}{k_B}\left(\frac{T_2 - T_1}{T_1}\right) = \boxed{31 \text{ J}} \text{ (positive!)}$$

(c) Immediately after the pull, what force must you exert on the piston to hold it in position (with the helium enclosed in a region that is 20 + 12 = 32 cm long)?



$P_2 V_2^{\gamma} = P_{air} V_1^{\gamma}$

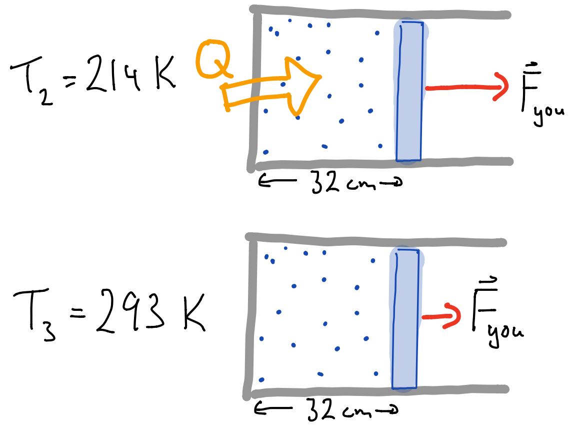
$$F_{you} = P_{air}A - P_2 A$$

$$= P_{air} A \left(1 - \left[\frac{V_1}{V_2}\right]^{\gamma}\right)$$

$$= (1 \times 10^5)(7.9 \times 10^{-3})\left(1 - \left[\frac{20}{32}\right]^{\frac{5}{3}}\right)$$

$$= \boxed{429 \text{ N}}$$

(d) You wait while the helium slowly returns to room temperature, holding the piston at its current location. After this wait, what force do you have to exert to hold the piston in position?



After waiting, the temperature and pressure inside have both increased. Since $T_3 = T_1$, we know

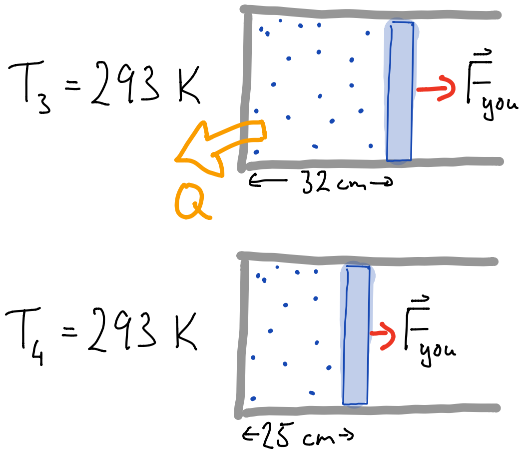
$$P_3 V_3 = P_{air} V_1$$

$$\rightarrow P_3 = P_{air}\left(\frac{V_1}{V_3}\right)$$

$$\therefore F_{you} = (P_{air} - P_3)A$$

$$= P_{air} A\left(1 - \frac{V_1}{V_3}\right) = \boxed{296\text{ N}}$$

(e) Next, you very *slowly* allow the piston to move back into the cylinder, stopping when the region helium gas is 25 cm long. What force must you exert to hold the piston at this position? What approximations did you make?



Assume helium is always in thermal equilibrium with the surroundings

$\rightarrow$ T is constant. (isothermal)

$$\therefore P_4 V_4 = P_3 V_3 = P_{air} V_1$$

$$P_4 = P_{air}\left(\frac{V_1}{V_4}\right)$$

$$\rightarrow F_{you} = (P_{air} - P_4)A$$

$$= P_{air} A\left(1 - \frac{V_1}{V_4}\right) = \boxed{158\text{ N}}$$